

Yoga AI Pose Detector

A Deep Learning Posture Assistant

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Created by:

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The Inspiration & Core Motivation

"Without the brilliant virtuoso yogini Kamini Panwar, this project would have never materialized. She has been the ultimate inspiration throughout this journey."

- Motivation: The deep internal desire to master as many Yoga poses as possible.
- The Hurdle: Receiving world-class guidance from Kamini for only one hour at the studio.
- The Solution: To build a robust tool that safely acts as a surrogate during home practice.
- Note: This tool is not intended to, and will never replace Kamini—the kindest, most enthusiastic, helpful, and sweet Yoga trainer I know.

Business Use Case & Value Proposition

- The Problem: High-quality yoga guidance is clustered in expensive studio sessions.
- Risk Factor: Practicing complex poses at home without expert eyes leads to injuries.
- Data Scarcity: Authoritative, granular Yoga alignment materials are hidden behind paywalls.
- The Vision: An accessible AI-powered virtual assistant providing 24/7 deterministic feedback.
- Outcome: Ensures safe, progressive home practice tracking 82 distinct fine-grained poses.

Under The Hood: Technology Stack

- Frontend Interface: React + Vite for a rapid, responsive single-page application.
- Backend Architecture: FastAPI using Async Python for rigorous endpoint validation.
- Computer Vision Foundation: MediaPipe extracting 33 high-fidelity 3D body landmarks.
- Classification Engine: Custom Deep Skeletal Multi-Layer Perceptron (81.3% Accuracy!).
- Generative Artificial Intelligence: Google Gemini 3.1 Pro strictly grounded via prompting.
- DevOps & Database: Docker for total containerization + MongoDB Atlas.

Demonstration: Platform Entry

Showcasing the seamless, user-friendly React landing interface.

[INSERT DEMO SCREENSHOT HERE]

Demonstration: Posture Feedback 1

Example of the Custom MLP accurately parsing joint angles against expected geometries.

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Demonstration: Posture Feedback 2

Highlighting Gemini's intelligent, layman-friendly instructor feedback.

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Conclusion & Future Trajectory

- Yoga AI bridges explicit geometric rules with the empathetic instruction of generative LLMs.
- Future Objective 1: Transitioning spatial features into Vision Transformers / EfficientNet.
- Future Objective 2: Full streaming video support for dynamic movement correction over time.
- Special Thanks to Kamini Panwar for her unparalleled dedication to her students.
- Special Thanks to Yuta, Raman, and the creators of the Yoga-82 dataset.

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